Overfitting Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Overfitting Intuition before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Overfitting Intuition trace and record the first step you can explain without help.

Use W2: Interpret a gap for **Overfitting** and model **capacity**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Overfitting Intuition explanation against the supplied facts before discussing it.

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Overfitting Intuition target to a trace and a justified conclusion.

Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

WORKBOOK · 6 OF 21

Solve the given case

Solve the supplied Overfitting Intuition case and show the deciding evidence.

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

> What does the **evidence** support?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> Rule A memorises six **training** book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Overfitting Intuition adds to Transparent Book Recommender and keep the evidence.

Avoid title-by-title patches; improve features or thresholds that transfer.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Select from evidence

Select from evidence. Apply the supplied Overfitting Intuition facts and keep your reasoning.

Choose among the three worked candidates and explain the **training** trap.

My reasoning and evidence

W2 Interpret a gap

> Interpret a gap. Apply the supplied Overfitting Intuition facts and keep your reasoning.

Does a **training**-validation gap alone prove **overfitting**?

My reasoning and evidence

W3 Propose a controlled change

> Propose a controlled change. Apply the supplied Overfitting Intuition facts and keep your reasoning.

Describe one fair next development comparison.

My reasoning and evidence

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Overfitting Intuition facts and keep your reasoning.

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

W5 Limit a diagnosis

> Limit a diagnosis. Apply the supplied Overfitting Intuition facts and keep your reasoning.

What extra **evidence** would you inspect before concluding a model memorized noise?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Overfitting Intuition

> A fictional project compares three book-length prediction models on the same development sets. Explain the labelled step before checking the changed condition.

A fictional project compares three book-length **prediction** models on the same development sets.

Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

AI ACADEMY / AI-2/16

Training and validation errors

	Training MAE pages	Validation MAE pages
Simple	5	6
Medium	2	3
Complex	0	7

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/16

Overfitting Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training comparison

Complex fits training perfectly with zero error.



2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.



3. Choose for development

Select Medium under the declared lowest-validation-error rule.



4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does zero **training** error justify choosing Complex in the original record?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional project compares three book-length **prediction** models on the same development sets.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Freeze the selection, then evaluate new reserved cases.

1. Final evaluation

2. **Training** comparison

3. Choose for development

4. Validation comparison

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Final evaluation Freeze the selection, then evaluate new reserved cases. A flexible model can fit quirks of its **training** set without predicting new examples well.

Here Complex's **training** success occurs alongside the worst validation error. The numbers are invented comparisons, not a claim that **complexity** always hurts.

Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Complex's validation error becomes 2 on a newly documented, fair development comparison. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The stated rule now selects Complex.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The stated rule now selects

> Complex. The rule compares validation errors; it does not ban complex models.

Repair a missing building block

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

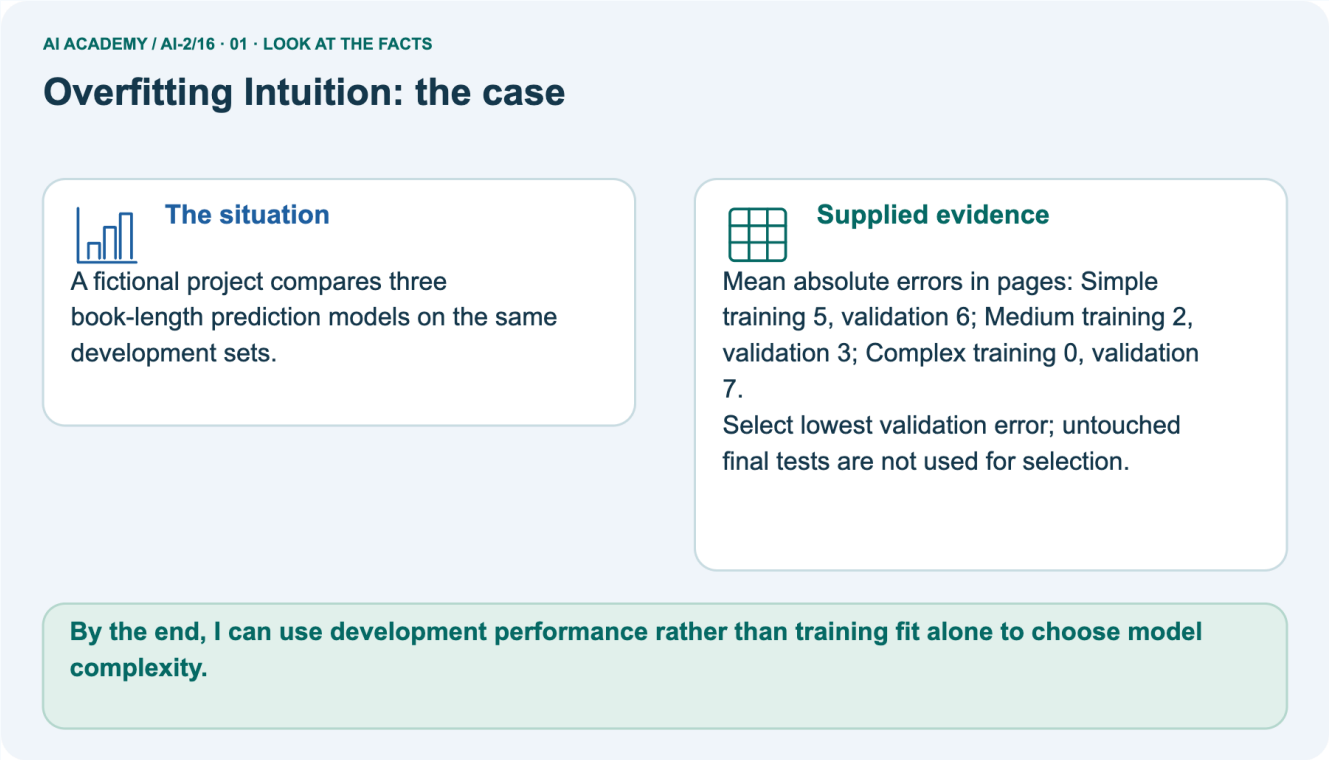
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Overfitting Intuition picture from its labels, then record a relevant limit.

01. Overfitting Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional project compares three book-length **prediction** models on the same development sets.

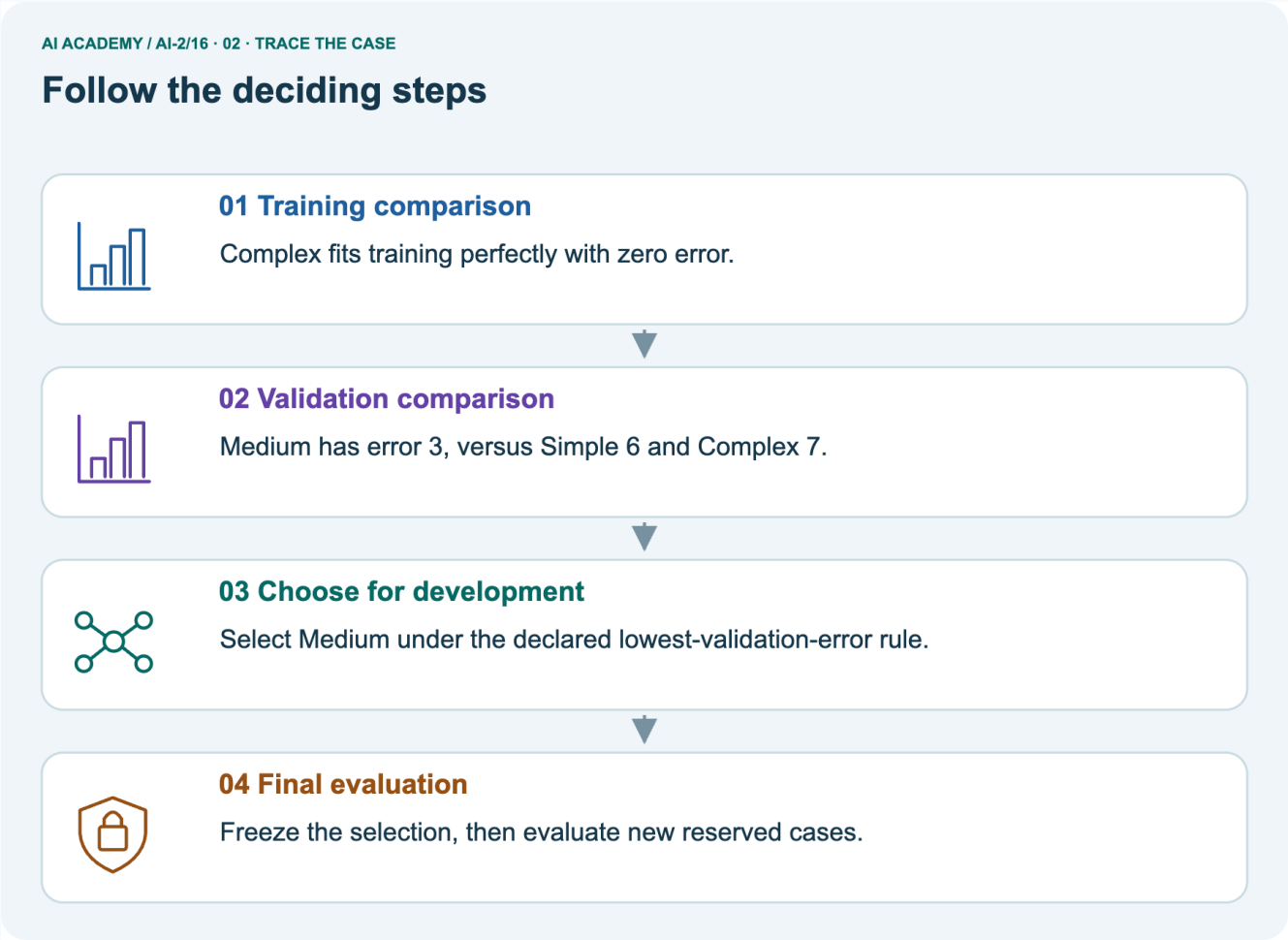
Supplied facts: Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



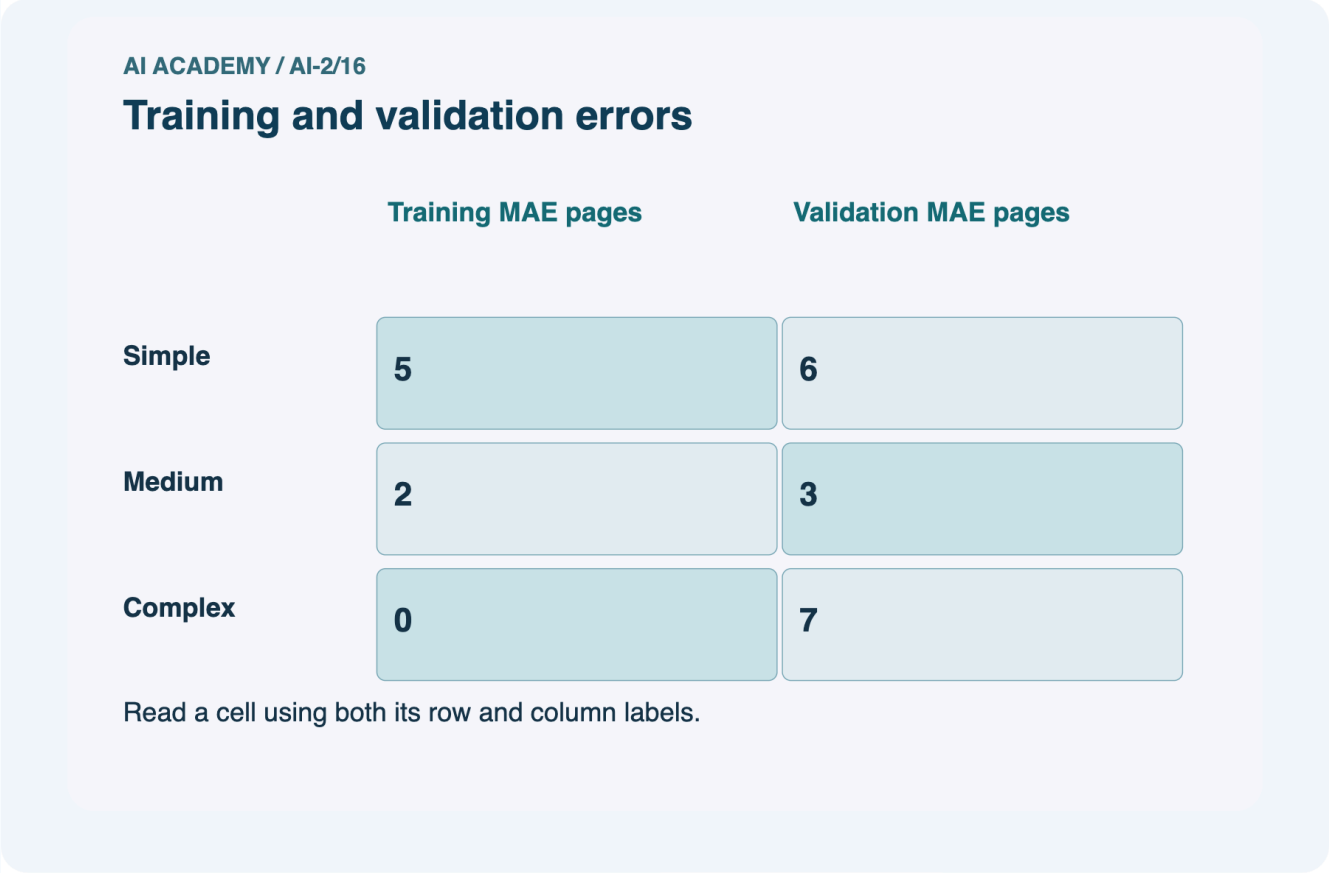
Follow the labelled picture with the matching explanation. Open the vector diagram

Training comparison: Complex fits **training** perfectly with zero error. → Validation comparison: Medium has error 3, versus Simple 6 and Complex 7. → Choose for development: Select Medium under the declared lowest-validation-error rule. → Final evaluation: Freeze the selection, then evaluate new reserved cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training and validation errors



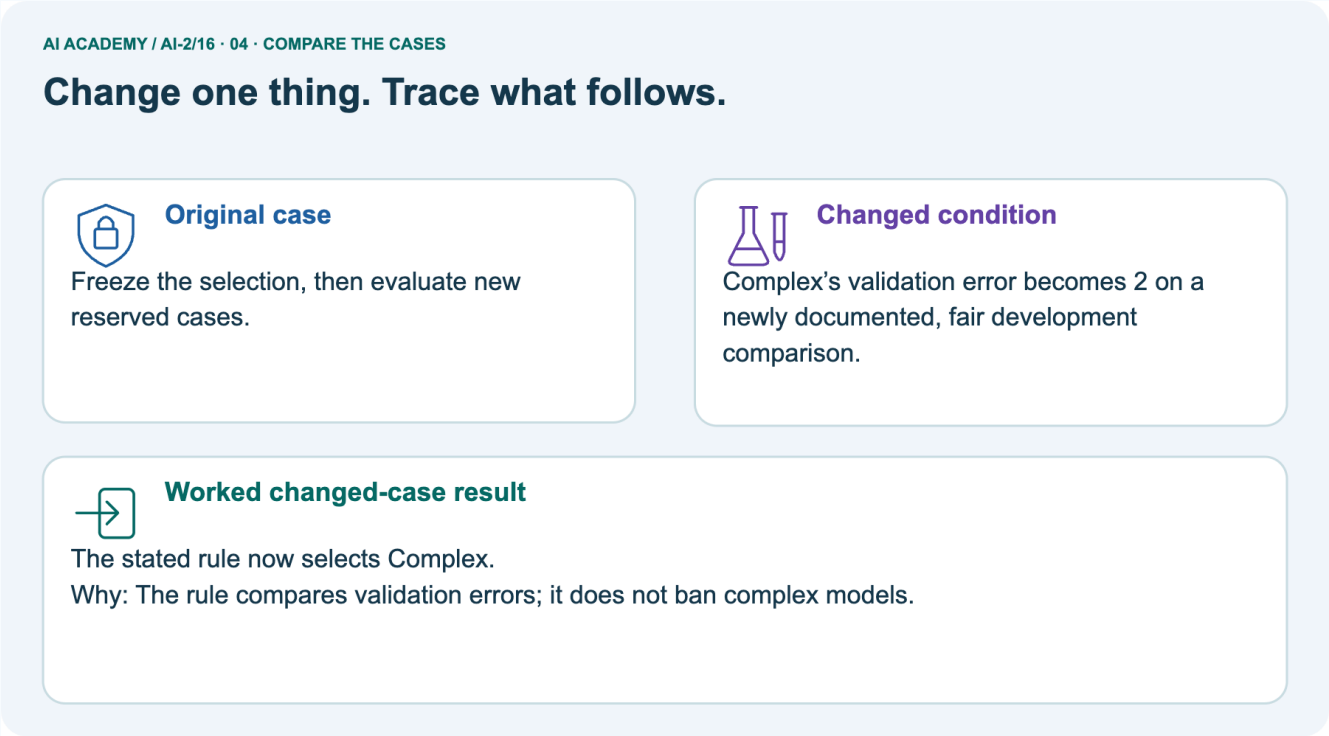
Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation errors. Simple: **Training** MAE pages = 5, Validation MAE pages = 6. Medium: **Training** MAE pages = 2, Validation MAE pages = 3. Complex: **Training** MAE pages = 0, Validation MAE pages = 7.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

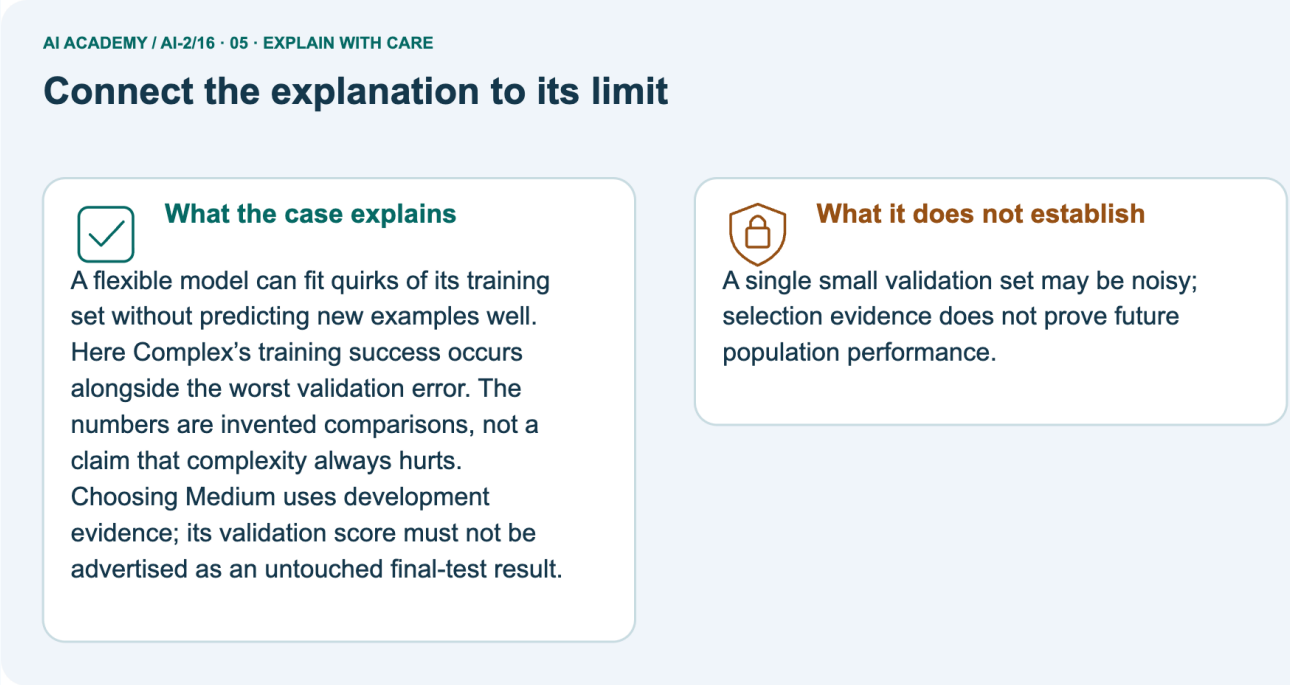
Original case: Freeze the selection, then evaluate new reserved cases. Changed condition: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Expected result: The stated rule now selects Complex. Reason: The rule compares validation errors; it does not ban complex models.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

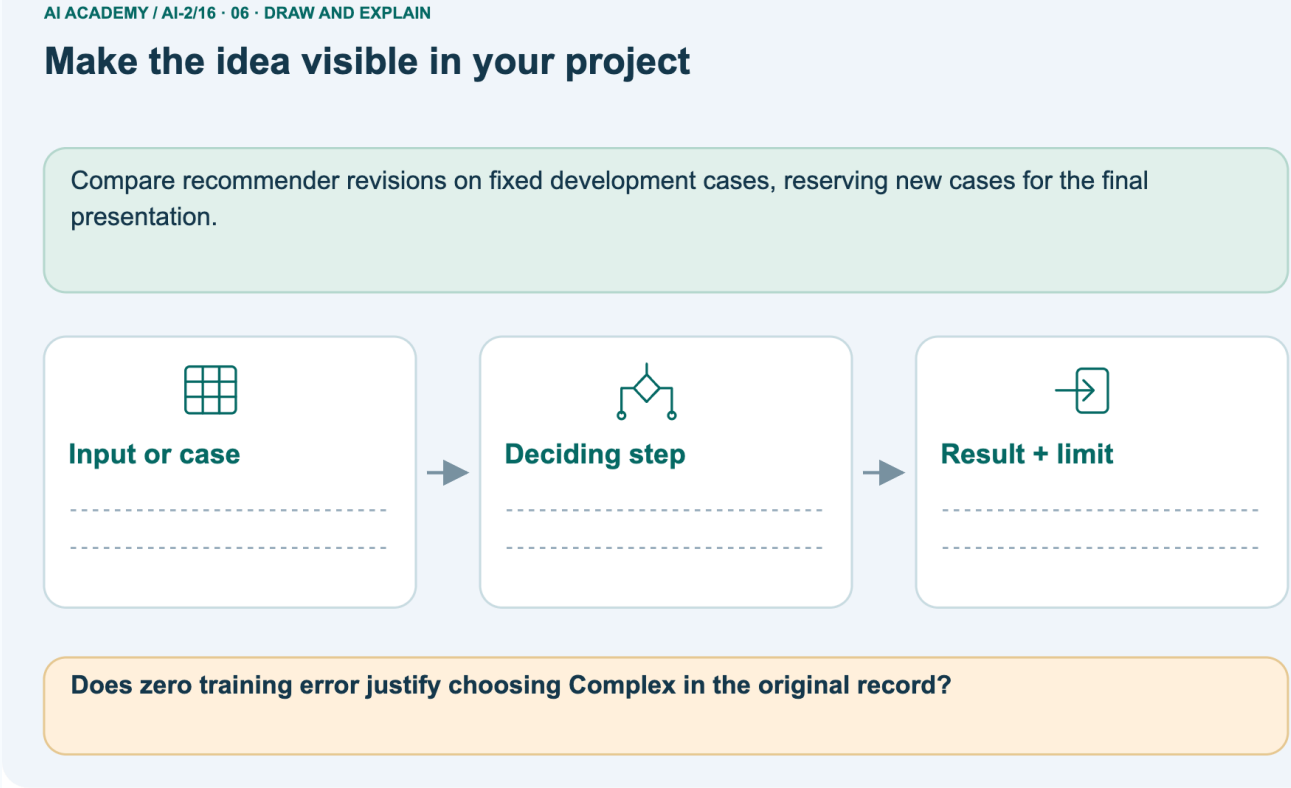
The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Limit: A single small validation set may be noisy; selection **evidence** does not prove future population performance.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation. Draw your **input**, deciding step, result and limit.

Supported question: Does zero **training** error justify choosing Complex in the original record?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.